FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Immunisation

Tutorial | Week 5 · Measles, vaccines & the newborn | Paediatrics | Dr Mawanda

Uganda's routine schedule (UNEPI, current MoH schedule 2025/26) now protects against 14 diseases. It includes a hepatitis B birth dose (introduced 2025), rotavirus, PCV and IPV in infancy, the R21 malaria vaccine at 6, 7, 8 and 18 months (107 high-burden districts from April 2025), measles–rubella at 9 and 18 months, yellow fever at 9 months and single-dose HPV for 10-year-old girls. Vaccinate at every contact, know the few true contraindications, and keep the cold chain at +2 to +8°C.

Must know before the ward round

1. Uganda routine schedule (MoH poster, current 2025/26): Birth BCG + OPV0 + HepB birth dose; 6, 10, 14 weeks OPV + DPT-HepB-Hib (pentavalent) + PCV + rotavirus (IPV at 6 and 14 weeks); 6 months malaria 1 + vitamin A; 7 and 8 months malaria 2 and 3; 9 months MR1 + yellow fever; 18 months MR2 + malaria 4 + vitamin A; 10-year-old girls HPV (1 dose).
2. Malaria vaccine (R21/Matrix-M): introduced April 2025 in 107 high-burden districts, 4 doses at 6, 7, 8 and 18 months (MoH media brief, January 2026). It adds to nets, IRS and prompt treatment; it does not replace them.
3. HPV: since 2025 Uganda gives a SINGLE dose to 10-year-old girls (Primary 4); immunocompromised girls (e.g. HIV) still get 2 doses. Hepatitis B birth dose introduced 2025 (75% coverage).
4. Routes and sites: BCG intradermal right upper arm (0.05 mL under 12 months); pentavalent IM left outer thigh; PCV IM right outer thigh; IPV IM right thigh 2.5 cm from PCV; OPV and rota oral; MR subcutaneous left upper arm; yellow fever right upper arm.
5. Cold chain: store all vaccines at +2 to +8°C at the health facility. NEVER freeze pentavalent, PCV, IPV, HepB, HPV, Td, rotavirus or malaria vaccine (shake test for freeze damage). OPV and MR may be kept frozen (–15 to –25°C) only at district/national stores.
6. VVM (vaccine vial monitor): stages 1–2 usable; stages 3–4 (inner square as dark as or darker than the circle) = discard. Reconstituted BCG, MR and yellow fever: discard after 6 hours or at the end of the session, whichever comes first.
7. True contraindications are few: anaphylaxis to a previous dose or component; live vaccines (BCG, yellow fever) in symptomatic HIV or severe immunosuppression; encephalopathy within 7 days of a pertussis-containing dose (no further pertussis); history of intussusception for rotavirus.
8. False contraindications (vaccinate!): mild illness or low-grade fever, diarrhoea, malnutrition, prematurity/low birth weight, antibiotics, breastfeeding, HIV exposure, jaundice, family history of convulsions, previous local reaction.
9. A severely ill child: UCG 2023 says admit, treat, and vaccinate at discharge. Minor illness is not a contraindication.
10. Catch-up: never restart a series; give every due antigen at the first contact; minimum interval between doses of the same vaccine is 4 weeks; give different live injectable vaccines on the same day or at least 4 weeks apart.
11. Anaphylaxis after vaccination: adrenaline 1 mg/mL (1:1000) 0.01 mg/kg = 0.01 mL/kg IM into the anterolateral thigh (max 0.5 mg), repeat every 5 minutes as needed; observe every vaccinee for 15–30 minutes.
12. AEFI: report serious events (death, hospitalisation, disability, clusters, events causing community concern) to the district within 24 hours on the AEFI form; classify as vaccine product-related, quality defect, immunisation error, immunisation anxiety, or coincidental (WHO 2018/2019).

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In a nutshell

Immunisation gives a child protective immunity by exposing the immune system to an antigen (live attenuated, inactivated, toxoid, subunit, conjugate or recombinant). It is the **most cost-effective child-survival intervention** there is: measles, neonatal tetanus, polio, Hib meningitis and pneumococcal and rotavirus disease have all fallen sharply in Uganda since vaccines were introduced.

Uganda's programme is run by **UNEPI** (Uganda National Expanded Programme on Immunisation) of the Ministry of Health. All routine vaccines are **free** at government and accredited private facilities. The schedule now protects against **14 diseases** and changed a lot in 2025: a **hepatitis B birth dose**, the **R21 malaria vaccine**, a **single-dose HPV** schedule and routine **yellow fever** at 9 months.

As a clerk you will meet immunisation every day: **every** history needs an immunisation status (from the **card**, not memory), every admitted child should leave **up to date**, and every outbreak (measles, polio) starts with a gap in coverage.

> **CLINICAL PEARL**
> The one thing never to forget: **every contact is a vaccination opportunity.** Check the card of every child you see (including the sick child on the ward and the sibling who came along), give what is due unless there is a **true** contraindication, and never send a child home "to come back when well" for a minor illness.

Definitions & classification

Term	Meaning
Immunisation / vaccination	Inducing immunity with a vaccine (active) or giving ready-made antibody (passive: immunoglobulin, e.g. HBIG, tetanus immunoglobulin)
Fully immunised child (Uganda)	Has received BCG, 3 doses of OPV, 3 of pentavalent, 3 of PCV, rotavirus doses, IPV and MR1 by 12 months (plus the newer antigens: malaria in eligible districts, MR2 at 18 months)
Zero-dose child	Has not received even the first pentavalent dose (DTP1); a marker of children the system misses
Dropout rate	(DTP1 − DTP3) ÷ DTP1 × 100. Over 10% suggests a programme problem
AEFI	Adverse event following immunisation: any untoward medical event after immunisation, not necessarily caused by the vaccine
Cold chain	The system of storing and transporting vaccines at recommended temperatures from manufacturer to child
VVM	Vaccine vial monitor: a heat-sensitive label that shows cumulative heat exposure
MDVP	WHO multi-dose vial policy: which opened vials may be kept for later sessions
Herd (community) immunity	Indirect protection of the unvaccinated when coverage is high (measles needs about **95%** with two doses)

Types of vaccine in the Uganda schedule

Type	Vaccines	Key point
Live attenuated	BCG, OPV, rotavirus, measles–rubella, yellow fever	Strong, long-lasting immunity from few doses; **avoid in severe immunosuppression** (BCG, YF) and pregnancy (MR, YF relative)
Inactivated (killed)	IPV	Safe in immunosuppression; needs boosters/several doses

Type	Vaccines	Key point
Toxoid	Tetanus and diphtheria (in pentavalent, Td)	Antibody against the toxin, not the bacterium
Subunit / recombinant	Hepatitis B, HPV, pertussis (whole-cell in Uganda's pentavalent)	Safe; adjuvant-containing; **freeze-sensitive**
Conjugate	Hib, PCV	Polysaccharide linked to protein so that **infants under 2** (who respond poorly to polysaccharide) mount T-cell-dependent memory
Protein/particle + adjuvant	Malaria R21/Matrix-M (circumsporozoite protein)	Pre-erythrocytic: blocks liver-stage infection

Epidemiology & Uganda context

- **Coverage:** in 2025 most antigens exceeded **90%** coverage nationally (MoH media brief, January 2026). The number of **zero-dose children** fell from about **104,675** at the start of 2024 to **36,452** in 2025, concentrated in **Buganda and Busoga**, urban poor, island, refugee-hosting and nomadic communities.
- **Measles:** COVID-19 disruptions left many under-immunised children. **66 districts** reported confirmed measles outbreaks in 2025 (896 confirmed cases), worst in **Karamoja** where malnutrition raised deaths; outbreak response vaccination ran in 65 of them, and a national measles–rubella campaign followed.
- **Malaria vaccine:** Uganda's **R21** introduction in April 2025 was the **largest malaria vaccine introduction worldwide**. By January 2026 over 1 million children had received dose 1, but only **67% dose 2 and 47% dose 3**: completion of a 4-dose schedule spread over 12 months is the challenge.
- **Hepatitis B:** about 6% of Ugandans have chronic viral hepatitis; the **birth dose** (2025) targets mother-to-child transmission, which pentavalent at 6 weeks comes too late to prevent.
- **Yellow fever:** Uganda is a high-risk country. Mass campaigns covered over 30 million people (ages 1–60) by 2025; protection is now maintained by **routine YF at 9 months**.
- **Polio:** Uganda is polio-free, but **circulating vaccine-derived poliovirus** outbreaks in the region keep surveillance for **acute flaccid paralysis (AFP)** essential.

> **UGANDA CONTEXT**
> Real-life barriers: vaccine and **fridge stock-outs**, **solar fridges** that fail on cloudy days, outreach sessions cancelled for lack of fuel, caregivers who lose the **child health card**, **misinformation** (rising since COVID-19), "**vaccines are only on Thursdays**" (missed opportunities), and private clinics that do not report doses. MoH responses include 1,076 new vaccine fridges, solarising 250 facilities, VHT follow-up of defaulters and targeted support to 57 "equity reference group" districts.

Principles: how vaccines work and why the schedule looks like this

- **Primary immune response:** antigen → antigen-presenting cells → T-helper cells → B cells → IgM then IgG and **memory cells**. Repeated doses (6, 10, 14 weeks) build higher-affinity, longer-lasting antibody. A **minimum interval of 4 weeks** lets each response mature; shorter intervals reduce the response.
- **Why start at 6 weeks?** Maternal IgG protects partly in the first months but **interferes** with some vaccines (especially **measles**, hence 9 months in high-transmission settings). DTP, Hib, PCV, rota and IPV work well from 6 weeks and the diseases strike young infants hard.
- **Why give some vaccines at birth? BCG** (TB meningitis and miliary TB in infancy), **OPV0** (gut immunity before exposure) and **HepB birth dose** (blocks perinatal HBV transmission; it must be given within 24 hours of birth to work best, ideally within the first day, at most the first 7 days per UCG).

- **Why two doses of MR?** About 85% of children seroconvert after a 9-month dose; the **18-month** dose catches the 15% who did not.
- **Why conjugate vaccines?** Children under 2 cannot respond to plain polysaccharide capsules (Hib, pneumococcus); conjugation to a protein recruits T-cell help.
- **Live vaccines** replicate, so in **severe immunosuppression** they can cause disease (disseminated BCG, YF viscerotropic disease). **Inactivated** vaccines cannot, but may work less well.
- **Malaria vaccine (R21):** a circumsporozoite protein particle with Matrix-M adjuvant. It targets **sporozoites** before they reach the liver. Three monthly priming doses from 5–6 months, then a booster 12 months after dose 3, give about **75% protection over the first year in seasonal settings** in the phase 3 trial (lower in perennial transmission). It reduces but does not abolish malaria, so **nets and prompt testing and treatment remain essential**.

Clinical acumen

Focused history

Ask	Why it matters
Look at the card (child health card / Mother–Child Health Passport): every antigen and date	Memory is unreliable; the card defines what is due. No card: check the register/BCG scar and give age-appropriate doses
Age in **weeks/months** (date of birth)	Decides what is due and whether catch-up is needed
Current illness: danger signs? fever?	Severe illness → admit and vaccinate at discharge; mild illness → vaccinate today
Previous reaction to a vaccine (what, when, how severe)	Anaphylaxis = contraindication to that vaccine; local reaction or fever = not a contraindication
Immunosuppression: HIV status (symptomatic?), steroids (≥2 mg/kg/day or ≥20 mg/day for ≥14 days), chemotherapy, SCID in family	Live vaccines (BCG, YF; also MR in severe immunosuppression)
History of intussusception	Contraindication to rotavirus vaccine
Encephalopathy, uncontrolled seizures or progressive neurological disease	Pertussis vaccine: contraindicated after encephalopathy within 7 days; defer in progressive/unstable neurological disease
Blood products or immunoglobulin in the past months	Can blunt MR response (measles vaccine needs an interval after immunoglobulin)
Birth weight and gestation	Preterm babies are vaccinated by **chronological** age
Mother's HBsAg and HIV status	HepB birth dose urgency; HEI care
Measles exposure or outbreak in the area	Early MR dose from 6 months
District of residence	Malaria vaccine eligibility (107 districts)

Examination

- **BCG scar** on the right upper arm (absent in a child with a documented dose is not a reason to repeat BCG under current WHO guidance).
- **Weight and Z-scores** (an immunisation visit is also a **growth-monitoring** visit), MUAC from 6 months.
- **Temperature**, danger signs (IMCI).
- Signs of **immunosuppression** (thrush, wasting, lymphadenopathy).
- Injection sites for **local reactions** or abscesses from previous doses.

Red flags

RED FLAG

- **Anaphylaxis** within minutes of a vaccine: stridor, wheeze, hypotension, urticaria, collapse. Adrenaline IM now.
- **Collapse, pallor and floppiness** (hypotonic–hyporesponsive episode) within 48 h of DTP: usually recovers, but assess for sepsis and hypoglycaemia.
- **Seizure** after vaccination: often a coincidental or febrile seizure, but exclude meningitis.
- **Acute flaccid paralysis** in any child under 15: notify as possible polio (AFP surveillance) and send two stool samples.
- **Fever with rash** (measles) in an unvaccinated child: isolate, notify, give vitamin A, check contacts.
- **Cluster** of abscesses or deaths after one session: suspect an **immunisation error** (contaminated reconstituted vial): stop the session and report at once.
- **Fridge alarm or frozen vaccine**: stop using the affected stock until assessed (shake test).

Thinking like a clinician

- **"Sick child, so no vaccines."** Only if the child is **severely** ill (admit, vaccinate at discharge). Cough, cold, diarrhoea, low-grade fever, antibiotics and malnutrition are **not** contraindications: the malnourished child is the one who most needs measles vaccine.
- **No card?** Don't restart. Ask, check the register, look for a BCG scar, and if still unknown, **assume unvaccinated** and give the age-appropriate doses: extra doses are safe.
- **Late for a dose?** Give it now; **never restart the series**. Plan the rest at minimum 4-week intervals.
- **HIV-exposed infant?** Vaccinate fully on schedule. Only **symptomatic HIV** changes things (no late BCG, no YF). MR is given even in symptomatic HIV because measles is so dangerous.
- **Premature baby?** Vaccinate by **chronological age**, not corrected age. BCG and OPV0 at birth or discharge; HepB birth dose for babies under 2 kg does not count towards the series.
- **Measles outbreak?** Give MR from **6 months** ("MR0"); it does not count, so repeat at 9 and 18 months.
- **A child who reacted badly last time**: ask exactly what happened. Fever, crying or a sore thigh is expected; **anaphylaxis** or **encephalopathy** changes management.

Clinical workup

For immunisation, the "workup" is the **immunisation encounter** and the **assessment of a possible AEFI**.

Step 0: First 5 minutes

1. **Triage** the child (ETAT/IMCI): any emergency or danger sign? Treat and admit; vaccinate at discharge.
2. Make sure **emergency drugs are ready before the session**: adrenaline 1 mg/mL, 1 mL syringes, bag-valve-mask, oxygen if available, IV fluids.
3. For a suspected **AEFI**: ABC first. If anaphylaxis → adrenaline 0.01 mL/kg of 1:1000 IM (max 0.5 mL), call for help, lie flat (legs raised), oxygen, IV fluids 10–20 mL/kg if shocked.

Step 1: Focused history

- [ ] Card: antigens and dates. *Defines what is due.*
- [ ] Age now. *Due doses and catch-up plan.*
- [ ] Illness today and severity. *Vaccinate today vs at discharge.*
- [ ] Previous reactions. *True vs false contraindication.*

- [ ] Immunosuppression, HIV status, steroids. *Live vaccines.*
- [ ] Intussusception history. *Rotavirus.*
- [ ] Neurological history. *Pertussis-containing vaccine.*
- [ ] District (malaria vaccine), outbreaks (MR0). *Extra or early doses.*

Step 2: Focused examination

- General: danger signs, temperature, weight and Z-scores, MUAC.
- BCG scar; previous injection sites; signs of immunosuppression.
- For an AEFI: vital signs, skin (urticaria, rash), airway (stridor), chest (wheeze), neurological state, injection site (swelling, abscess, lymphadenitis).

Step 3: Bedside checks (on the vaccine, not the child)

Check	What to look for
Fridge thermometer / Fridge-tag	+2 to +8°C; record **twice daily**, including weekends
Expiry date	Never use expired vaccine or diluent
VVM	Inner square lighter than outer circle = usable; same or darker = **discard**
Shake test (freeze-sensitive vaccines)	Suspect freezing → shake suspect and frozen control vials; if the suspect sediments as fast as the control = freeze-damaged, discard
Diluent	Same manufacturer, **pre-cooled**; never water for injection or another drug's diluent
Reconstitution time	BCG, MR, YF: write the time on the vial; discard after **6 hours** or session end
Label	No label = discard

Step 4: Laboratory & imaging

Routine immunisation needs no tests. Investigations matter for **AEFI** and **vaccine-preventable disease (VPD) surveillance**.

Test	When	Expected / abnormal finding	How it changes management	Availability
RBS	Floppy, unresponsive or seizing child after vaccination	Hypoglycaemia	Treat (10% dextrose 2–5 mL/kg); points to a coincidental cause	Every level (glucometer)
Malaria RDT / blood smear	Fever after vaccination	Positive	Coincidental malaria: treat it; do not blame the vaccine	HC II and above
FBC, blood culture	High fever, toxic child, injection-site abscess cluster	Bacteraemia (e.g. *Staph aureus*, Gram-negatives)	Suggests contaminated vial (immunisation error): investigate the session	Hospital
Lumbar puncture	Seizure with fever, altered consciousness after vaccine	Meningitis	Coincidental meningitis: treat	Hospital
Abdominal ultrasound	Vomiting, red-currant stool, abdominal mass within 7 days of rotavirus vaccine	Intussusception (target sign)	Reduction (air/hydrostatic) or surgery; no further rota doses	Regional/national referral
Stool samples ×2 (24 h apart, within 14 days of onset)	Acute flaccid paralysis in a child <15	Wild or vaccine-derived poliovirus	Polio response; mandatory notification	Collected at any facility, sent to UVRI via district

Test	When	Expected / abnormal finding	How it changes management	Availability
Measles IgM (serum/DBS)	Fever + maculopapular rash + cough/coryza/conjunctivitis	Positive IgM	Confirms measles (and rubella) outbreak; triggers response	Via district to UVRI
CXR, lymph node aspirate	Large BCG node or abscess, especially in HIV	BCG lymphadenitis/disseminated BCG	Needle aspiration; TB-type treatment in disseminated BCGosis	Hospital

Step 5: Putting it together

Model one-liner: "Baby N, 8 months, weight 7.6 kg (WAZ −0.9), HIV-unexposed, exclusively breastfed to 6 months, from Wakiso (malaria vaccine district), presents for immunisation; her card shows BCG, OPV0, HepB-BD, and 6-, 10- and 14-week doses, but no malaria vaccine and no vitamin A."

Worked example: reading the card and planning catch-up

- Due today (8 months): **malaria vaccine dose 1** (she missed the 6-month start; give now) + **vitamin A 100,000 IU**.
- Next: **malaria dose 2 in 4 weeks** (9 months, together with **MR1 + yellow fever**), **malaria dose 3** 4 weeks later (10 months), then **malaria dose 4** 12 months after dose 3, plus **MR2 at 18 months** and vitamin A every 6 months.
- Write the next date on the card and explain it to the mother.

Worked example: a 14-month-old with no doses since 6 weeks

- Give today: **OPV2, pentavalent 2, PCV2, IPV2, MR1, yellow fever**, plus rotavirus only if your national guidance allows doses at this age (WHO: rota catch-up can continue to 24 months where rotavirus mortality is high). Then OPV3, penta 3 and PCV3 after 4 weeks, MR2 at least 4 weeks after MR1 (or at 18 months). Malaria vaccine per district policy (the target is children under 2).

Differential diagnosis

Is it the vaccine or a coincidence? Most events after vaccination are **coincidental**.

Event after vaccination	Vaccine-related features	Coincidental / other causes to exclude	Key investigation
Fever	Within 24–48 h of pentavalent/PCV; 5–12 days after MR or YF; low-grade, settles	**Malaria**, pneumonia, UTI, otitis, meningitis	Malaria RDT, examine fully
Seizure	Febrile seizure 24–72 h after DTP or 7–10 days after MR; brief and generalised	**Meningitis**, cerebral malaria, hypoglycaemia, epilepsy	RBS, RDT, LP if indicated
Floppy, pale, unresponsive	HHE within 48 h of DTP; recovers spontaneously	Sepsis, hypoglycaemia, cardiac disease	RBS, vitals, observe
Swelling at site	Local reaction: red, sore, small, 1–3 days	**Abscess** (hot, fluctuant, days later: suggests non-sterile technique); "sterile abscess" from adjuvant	Aspirate if fluctuant
Axillary lymph node	**BCG lymphadenitis**: ipsilateral (right) axillary node, weeks–months after BCG	TB adenitis, bacterial adenitis, HIV lymphadenopathy	HIV test, aspirate

Event after vaccination	Vaccine-related features	Coincidental / other causes to exclude	Key investigation
Vomiting, blood in stool	**Intussusception** within 7 days of rotavirus (very rare, about 1–6 per 100,000)	Dysentery, coincidental intussusception (peak 4–10 months)	Ultrasound
Paralysis	**VAPP** after OPV (about 1 per 2.7 million doses)	Guillain–Barré, traumatic neuritis from a buttock injection, transverse myelitis, wild/cVDPV polio	Stool samples (AFP surveillance)
Rash	MR rash 7–12 days after; urticaria within hours (allergy)	Measles infection, drug rash	History of timing

Investigations

Test	What you are looking for	Why
Card, register, BCG scar	Doses received	Decide what is due
HIV status (child/mother)	Symptomatic HIV	Live vaccine decisions; HEI care
AEFI workup (RBS, RDT, FBC, culture, LP as indicated)	Coincidental illness, contamination	Causality assessment
AFP stool samples	Poliovirus	Polio eradication surveillance
Measles IgM	Measles/rubella	Outbreak confirmation
Fridge temperature log, VVM, shake test	Cold chain failure	Vaccine potency

Management

A. The current Uganda routine schedule (MoH UNEPI schedule, 2025/26)

Age	Vaccines	Dose and route	Site
Birth (or first contact)	**BCG**	0.05 mL **intradermal** (0–11 months); 0.1 mL from 12 months	**Right** upper arm (deltoid insertion)
	OPV0	2 drops oral	Mouth (ideally within the first 2 weeks)
	Hepatitis B birth dose (new 2025)	0.5 mL **IM**	Outer thigh; give **within 24 hours** of birth (UCG: within the first 7 days)
6 weeks	**OPV1**	2 drops oral	Mouth
	DPT-HepB-Hib 1 (pentavalent)	0.5 mL **IM**	**Left** outer thigh
	PCV1	0.5 mL IM	**Right** outer thigh
	Rotavirus 1	Oral (volume per product)	Slowly on the inner cheek
	IPV1 (UCG 2023)	0.5 mL IM	Right outer thigh, **2.5 cm** from PCV
10 weeks	**OPV2, Penta 2, PCV2, Rota 2**	As above	As above

Age	Vaccines	Dose and route	Site
14 weeks	**OPV3, Penta 3, PCV3, Rota 3** (current poster), **IPV2** (UCG 2023)	As above	As above
6 months	**Malaria vaccine 1** (R21) + **vitamin A**	0.5 mL IM; vitamin A 100,000 IU oral	Thigh (or deltoid per product guidance)
7 months	**Malaria vaccine 2**	0.5 mL IM	
8 months	**Malaria vaccine 3**	0.5 mL IM	
9 months	**Measles–rubella 1** + **yellow fever**	MR 0.5 mL **subcutaneous**; YF 0.5 mL (SC per UCG table)	MR **left** upper arm; YF **right** upper arm
18 months	**MR2** + **malaria vaccine 4** + **vitamin A**	As above; vitamin A 200,000 IU	
10-year-old girls (Primary 4, or age 10 out of school)	**HPV**	**1 dose** 0.5 mL IM (2 doses, 6 months apart, if immunocompromised)	Left upper arm
Women 15–49 years	**Td** (tetanus–diphtheria) 5 doses	0.5 mL IM	Upper arm

UGANDA CONTEXT

The schedule has changed faster than the textbooks. **UCG 2023** (adapted from the 2022 UNEPI schedule) still shows **2 rotavirus doses (6 and 10 weeks)**, **2-dose HPV** and no HepB birth dose or malaria vaccine. The **MoH schedule poster (2025/26)** shows **rotavirus at 6, 10 and 14 weeks**, a **HepB birth dose**, **malaria vaccine** at 6, 7, 8 and 18 months and **one HPV dose**. Follow the current MoH poster and the product in your fridge (the number of rotavirus doses depends on the brand). VERIFY the IPV schedule (UCG 2023: 6 and 14 weeks) with your facility.

Td schedule for women of childbearing age (15–49 years): Td1 at first contact; Td2 at least 4 weeks later (3 years' protection); Td3 6 months after Td2 (5 years); Td4 1 year after Td3 (10 years); Td5 1 year after Td4 (lifelong/30 years). This prevents **neonatal tetanus**.

Vitamin A supplementation (with immunisation contacts): 6–11 months **100,000 IU**; 12–59 months **200,000 IU** every 6 months.

B. Running a safe immunisation session (step by step)

1. **Prepare**: check fridge temperature, take vaccines in a cold box/carrier with conditioned ice packs; **sponge method** (vials in the foam pad) during the session; emergency tray ready.
2. **Screen** each child: card, age, illness, previous reactions, immunosuppression (questions above).
3. **Counsel** the caregiver: which vaccines, which diseases, expected side-effects, when to return; obtain verbal consent.
4. **Check the vial**: name, expiry, VVM, diluent, reconstitution time, not frozen.
5. **Prepare and give**: one **auto-disable (AD) syringe** per injection; correct route, site and dose; oral vaccines first (OPV, rota), then injections; never more than one injection in the same site; space thigh injections 2.5 cm apart.
6. **Dispose** of sharps immediately in a **safety box** (no recapping).
7. **Record** on the card, in the child register and tally sheet; write the **next date** on the card.
8. **Observe** for **15–30 minutes**.
9. **Advise**: paracetamol if fever, do not apply anything to injection sites, BCG ulcer is normal, return if seizures, severe rash, breathing difficulty or very high fever.

C. Contraindications and precautions

Situation	Vaccines affected	Action
Anaphylaxis to a previous dose or a component	That vaccine	**Do not give**
Symptomatic HIV / severe immunosuppression (SCID, chemotherapy, high-dose steroids)	**BCG** (if missed at birth), **yellow fever**; (MR and rota: specialist advice in SCID/severe immunosuppression)	Withhold live vaccines; **MR still given in HIV** (UCG)
Encephalopathy within 7 days of a pertussis-containing dose, not explained by another cause	Pentavalent (pertussis component)	No further pertussis; continue with DT-containing alternatives where available
Progressive or unstable neurological disorder	Pertussis-containing vaccines	Defer until stable
History of intussusception, SCID	Rotavirus	Do not give
Severe illness (needing admission)	All	Admit, treat, **vaccinate at discharge** (UCG 2023)
Pregnancy	MR, yellow fever (unless high risk), HPV	Defer; Td is recommended in pregnancy
Recent immunoglobulin or blood	MR	Defer (interval depends on the product)
Infant under 6 months	Yellow fever	Never under 6 months (encephalitis risk)

D. False contraindications: vaccinate

- Minor illness: cough, cold, diarrhoea, **low-grade fever**.
- **Malnutrition** (a reason to prioritise vaccination).
- **Prematurity, low birth weight** (use chronological age).
- On **antibiotics**, recovering from illness.
- **Breastfeeding**; mother pregnant.
- **HIV exposure** or asymptomatic HIV.
- **Neonatal jaundice**.
- **Family history** of convulsions, SIDS or vaccine reactions.
- Previous **local reaction** or mild fever after a dose.
- **Stable neurological conditions** (cerebral palsy, Down syndrome), eczema, asthma, allergies not to vaccine components.
- **Recent or upcoming surgery**; being an orphan; not having a card.

E. Managing AEFIs

Event	Management	Drug	Dose	Route	Frequency / max
Anaphylaxis	ABC, call for help, lie flat, legs raised, oxygen	**Adrenaline 1 mg/mL (1:1000)**	**0.01 mg/kg = 0.01 mL/kg** (under 6 years: 0.15 mL is a common practical dose; 6–12 years 0.3 mL; >12 years 0.5 mL)	**IM** anterolateral thigh	Repeat every **5 minutes** if no improvement; max **0.5 mg** per dose
	Shock	Normal saline/Ringer's	**10–20 mL/kg** bolus	IV/IO	Reassess after each bolus

Event	Management	Drug	Dose	Route	Frequency / max
	After stabilisation (adjuncts, not first line)	Chlorphenamine; hydrocortisone	Chlorphenamine 0.1 mg/kg (max 4 mg; 2.5 mg for 6 months–6 years); hydrocortisone 4 mg/kg (max 200 mg)	IV/IM/oral	Observe at least **6–12 hours**; refer
Fever, pain	Reassure, fluids, continue breastfeeding	**Paracetamol**	**15 mg/kg**	Oral	6-hourly; max **60 mg/kg/day**
Febrile seizure	Recovery position, ABC, time it	Diazepam (if >5 min)	**0.5 mg/kg rectal** (max 10 mg) or 0.3 mg/kg IV slowly	PR/IV	Once, may repeat after 10 min; look for meningitis/malaria
HHE	Observe, ABC, RBS	–	–	–	Usually resolves; not a contraindication to further doses in most guidance
BCG ulcer	Normal: keep clean and dry, **do not cover or apply anything**	–	–	–	Heals in about 2–3 months with a scar
BCG lymphadenitis	Non-suppurative: observe (resolves); suppurative/fluctuant: **needle aspiration**	–	–	–	Antibiotics do not help; test for HIV
Injection abscess	Incision and drainage; antibiotics if cellulitis	Flucloxacillin or cloxacillin	25 mg/kg	Oral	6-hourly, 5–7 days (max 500 mg per dose)

DRUG DOSING

Adrenaline is the only first-line drug for anaphylaxis. Antihistamines and steroids are adjuncts and never replace it. Give it **IM in the thigh**, not subcutaneously, and not the IV route in a clinic setting.

Reporting an AEFI (Uganda):

1. Treat the child.
2. Fill in the **AEFI case investigation/reporting form**; serious AEFIs (death, hospitalisation, disability, clusters, events causing community concern) go to the **district** (DHO/EPI focal person) **within 24 hours**; the district informs UNEPI and the **National Drug Authority (NDA)** pharmacovigilance system.
3. **Keep the vial** and syringes from the session (do not discard) for investigation.
4. The investigation team classifies causality (WHO 2018/2019): **vaccine product-related**, **vaccine quality defect**, **immunisation error-related** (e.g. contaminated vial, wrong diluent, wrong route), **immunisation anxiety-related** (fainting, hyperventilation, especially in adolescents during HPV campaigns) or **coincidental**.
5. Communicate honestly with the family and community; rumours spread faster than facts.

F. Catch-up immunisation

1. **Never restart** a series; count every documented dose.
2. Give **all due antigens at the first contact**, at different sites.
3. Respect **minimum intervals**: 4 weeks between doses of OPV, pentavalent, PCV, IPV, rota, MR and malaria primary doses.
4. **Live injectable vaccines** (MR, YF, BCG) can be given the **same day**; otherwise separate by **at least 4 weeks**.

5. **Age limits** to know (WHO position papers; VERIFY local practice): BCG for unvaccinated children (Uganda gives 0.1 mL from 12 months); **Hib** not usually needed after 5 years; **PCV** catch-up for children under 5; **rotavirus** started as early as possible and generally not given after 24 months; MR for any unvaccinated child; malaria vaccine for children in the eligible age group in eligible districts.
6. Uganda took part in the **"Big Catch-Up"** (WHO/UNICEF/Gavi, 2023–2025) for children who missed doses during COVID-19.

G. Special situations

- **Measles exposure or outbreak**: MR from **6 months** (does not count; repeat at 9 and 18 months); MR within **72 hours** of exposure can prevent disease in unvaccinated contacts; give **vitamin A** to every child with measles (see *Measles*).
- **Baby of an HBsAg-positive mother**: HepB birth dose **within 24 hours** (plus HBIG where available), then pentavalent at 6, 10 and 14 weeks; test the infant for HBsAg and anti-HBs at 9–12 months where possible.
- **Preterm/LBW**: chronological age; BCG, OPV0, HepB at birth/discharge.
- **HIV**: see *HIV/AIDS in Children*; HPV 2 doses.
- **Tetanus-prone wound**: Td and tetanus immunoglobulin per UCG (TIG 75 IU under 5 years, 125 IU 5–10 years, 250 IU over 10 years, IM).

Complications

- **Local**: pain, swelling, sterile abscess, BCG ulcer/lymphadenitis, injection abscess (technique), sciatic nerve injury (buttock injections; never inject the buttock in infants).
- **Systemic**: fever, febrile seizures, HHE, rash (MR), arthralgia (rubella component in adolescents and women).
- **Severe and rare**: anaphylaxis (about 1 per million doses), intussusception (rotavirus), **VAPP** and **cVDPV** (OPV, the reason for adding IPV and switching to bivalent OPV), disseminated BCG (immunosuppression), YF-associated neurotropic/viscerotropic disease (rare; contraindicated under 6 months).
- **Programme failures**: cold-chain breaks (impotent vaccine and outbreaks in "vaccinated" children), **immunisation errors** (toxic shock from contaminated reconstituted vials, wrong diluent: insulin or muscle relaxant deaths reported globally), missed opportunities and dropout, vaccine hesitancy.

Ward-round & exam pearls

Model case presentation (child):

> "Mukasa is a 15-month-old boy from Kayunga, weight 8.1 kg (WAZ −2.4), length 74 cm (LAZ −2.1), WLZ −1.6, MUAC 12.8 cm, admitted with measles complicated by pneumonia. He is HIV-unexposed, was exclusively breastfed to 6 months and now eats family food and breastfeeds. His card shows BCG (scar present), OPV0, and 6-week doses only; he has received no doses since because his mother moved and 'lost the clinic'. He has had no vitamin A. So he is a partially immunised child who missed MR1 at 9 months, and this admission is a missed-opportunity story. Plan: treat the pneumonia, vitamin A on days 1, 2 and (eye signs) 15, notify measles, vaccinate contacts and siblings, and before discharge give his catch-up doses (OPV, pentavalent, PCV, IPV, yellow fever) with a written plan for the rest. He should still receive MR (it protects against rubella, and clinical measles is not always measles)."

Likely viva questions (with answers)

1. *What is given at birth in Uganda?* BCG, OPV0 and hepatitis B birth dose.
2. *Why is measles vaccine given at 9 months and not at 6 weeks?* Maternal antibody neutralises the live vaccine; by 9 months most maternal antibody has gone while the risk of measles is high.

3. *Which vaccines must never be frozen?* Pentavalent, PCV, IPV, HepB, HPV, Td, rotavirus and malaria vaccine (adsorbed/protein vaccines).
4. *What does stage 3 on a VVM mean?* The inner square matches the outer circle: discard point; do not use.
5. *What is the malaria vaccine schedule in Uganda?* R21, 6, 7, 8 and 18 months, in 107 high-burden districts since April 2025.
6. *List four false contraindications.* Mild illness, malnutrition, prematurity, antibiotics (also breastfeeding, HIV exposure, family history of convulsions).
7. *How do you manage anaphylaxis after a vaccine?* Adrenaline 0.01 mL/kg of 1:1000 IM in the thigh (max 0.5 mL), repeat every 5 minutes, ABC, fluids, refer, report as an AEFI.

OSCE tips: in a **"counsel the mother"** station, use plain words for each disease, mention **expected** side-effects and **what to do**, give the **next date**, check understanding (teach-back), and invite questions about rumours. In a **"read this card"** station, state the child's age, list what is missing, what is due today and the catch-up plan.

How to present this tutorial

Edith presents this tutorial. Aim for **20–25 minutes** plus **10 minutes** of questions.

Suggested structure

1. **Hook (1 min):** "In 2025, 66 Ugandan districts had measles outbreaks. Why, when we have had a measles vaccine since the 1960s?" (Coverage gaps from COVID-19, zero-dose children, cross-border movement.)
2. **Objectives (30 s):** know the schedule; give vaccines safely; tell true from false contraindications; manage AEFIs; plan catch-up.
3. **Basic immunology (2 min):** live vs inactivated vs conjugate; why the 4-week interval; maternal antibody and measles timing.
4. **The schedule (5 min):** draw the **timeline** (Birth → 6 → 10 → 14 weeks → 6, 7, 8, 9 → 18 months → 10 years). Highlight what is **new**: HepB birth dose, malaria vaccine, single-dose HPV, rotavirus doses on the current poster. Bring the **MoH poster**.
5. **Routes and sites (2 min):** a quick table, or demonstrate on a doll (left thigh pentavalent, right thigh PCV + IPV 2.5 cm apart, right arm BCG, left arm MR).
6. **Cold chain (3 min):** +2 to +8°C, freeze-sensitive list, VVM pictures, shake test, MDVP and the 6-hour rule.
7. **Contraindications (3 min):** a "vaccinate or not?" quiz with 5 quick scenarios (fever 37.9°C; SAM; symptomatic HIV and BCG; previous anaphylaxis; preterm 34 weeks).
8. **AEFI (3 min):** classification, anaphylaxis drill with adrenaline dose, reporting pathway (district within 24 hours, NDA).
9. **Catch-up (2 min):** worked example of a card with gaps.
10. **Uganda programme (1 min):** coverage, zero-dose numbers, challenges.
11. **Take-home messages (30 s).**

Delivery tips

- Bring a **real child health card** (or a photo) and walk the group through it.
- Expect: *Why does Uganda give OPV and IPV? What is cVDPV? Is the malaria vaccine for all districts? Can you give MR to an HIV-positive child? What if a mother refuses?* (Explore reasons, correct misinformation respectfully, document, and keep the door open; see *Professionalism & Ethics*.)
- Say clearly which numbers come from the **MoH January 2026 media brief** and that UCG 2023 is older than the current schedule.

Mnemonics & memory aids

- **"BOH at birth"**: **B**CG, **O**PV0, **H**epB birth dose.
- **"OPPRI" at 6, 10, 14 weeks**: **O**PV, **P**entavalent, **P**CV, **R**ota, **I**PV (IPV at 6 and 14 weeks).
- **Malaria "6-7-8 … 18"**: three monthly doses, then a booster at 18 months.
- **"9 is MRY"**: **M**easles–**R**ubella + **Y**ellow fever at 9 months.
- **Left leg Penta, Right leg PCV** ("**P**enta **L**eft, **P**neumo **R**ight").
- **Never freeze "PHI-HTR-M"**: **P**enta/PCV, **H**epB, **I**PV, **H**PV, **T**d, **R**ota, **M**alaria.
- **VVM**: "square **lighter** than circle = **use**; square **matches or darker** = **discard**."
- **AEFI causes "PQ-EAC"**: **P**roduct, **Q**uality defect, **E**rror, **A**nxiety, **C**oincidental.

High-yield summary

Numbers & doses to know

Item	Value
Diseases covered by Uganda's programme	**14**
Birth	**BCG + OPV0 + HepB birth dose** (HepB-BD introduced **2025**)
6, 10, 14 weeks	**OPV, pentavalent, PCV, rotavirus** (+ IPV at 6 and 14 weeks)
Malaria vaccine (R21)	**6, 7, 8 and 18 months**; **107 districts**; introduced **April 2025**
9 months	**MR1 + yellow fever**
18 months	**MR2** (+ malaria 4, vitamin A)
HPV	Girls **10 years**, **1 dose** (2 if immunocompromised), since **2025**
Td (women 15–49)	**5 doses**: 0, 4 weeks, 6 months, 1 year, 1 year
BCG dose	**0.05 mL ID** under 12 months; 0.1 mL from 12 months; right upper arm
Pentavalent, PCV, IPV, HepB	**0.5 mL IM** (thigh in infants)
MR	**0.5 mL SC**, left upper arm
Storage	**+2 to +8°C**; record **twice daily**
Reconstituted BCG/MR/YF	Discard after **6 hours** or session end
Opened OPV, IPV, penta, PCV, Td vials (MDVP conditions met)	Up to **4 weeks** (28 days)
Minimum interval between doses	**4 weeks**
Observation after vaccination	**15–30 minutes**
Adrenaline (anaphylaxis)	**0.01 mL/kg of 1 mg/mL IM** (max 0.5 mL), repeat every **5 min**
Paracetamol for post-vaccine fever	**15 mg/kg** 6-hourly, max **60 mg/kg/day**
Vitamin A	6–11 months **100,000 IU**; 12–59 months **200,000 IU** every 6 months
Measles herd-immunity threshold	About **95%** two-dose coverage
AFP stool samples	**2 samples, 24 h apart, within 14 days** of paralysis onset
Serious AEFI report	To district **within 24 hours**

Item	Value
Zero-dose children	**104,675 (2024) → 36,452 (2025)**

Classic exam traps

- **"The child has a cold; come back next week."** Wrong: minor illness is a **false contraindication**; vaccinate today.
- **"The child is malnourished, so vaccines won't work; wait."** Wrong: malnourished children are at highest risk; vaccinate.
- **"Restart the series because the child is late."** Wrong: **never restart**; continue where they stopped.
- **"Correct for prematurity before vaccinating."** Wrong: use **chronological** age.
- **"Give BCG to a 2-year-old with symptomatic HIV who missed it."** Wrong: live BCG is contraindicated in symptomatic HIV.
- **"Withhold MR from an HIV-positive child."** Wrong: MR is given even in symptomatic HIV (UCG 2023).
- **"Store OPV in the freezer compartment of the HC III fridge."** At facility level all vaccines are kept at **+2 to +8°C**; freezing is for district/national stores only.
- **"Antihistamine first for anaphylaxis."** Wrong: **IM adrenaline first**.

Questions seniors ask

What does the Uganda schedule look like now? Birth BCG, OPV0, HepB birth dose; 6, 10, 14 weeks OPV, pentavalent, PCV, rotavirus (IPV at 6 and 14 weeks); malaria 6, 7, 8 months; 9 months MR and yellow fever; 18 months MR2 and malaria 4; HPV single dose at 10 years for girls.

Which vaccines are live, and why does it matter? BCG, OPV, rotavirus, MR and yellow fever; avoid BCG and YF in symptomatic HIV/severe immunosuppression, and MR and YF in pregnancy.

A mother says her child had fever and cried all night after the last pentavalent dose. Will you give the next one? Yes. Fever and irritability are expected minor reactions; give paracetamol advice. Only anaphylaxis or encephalopathy within 7 days would stop further pertussis doses.

How do you know a vaccine is still good? Fridge log within +2 to +8°C, not expired, VVM inner square lighter than the circle, not frozen (shake test for adsorbed vaccines), label intact, reconstituted vials within 6 hours.

A child has a seizure 8 days after MR. What do you think? Most likely a febrile seizure from the expected post-MR fever (days 5–12) or a coincidental illness; examine for meningitis and malaria, check glucose, treat, report as an AEFI and do not withhold MR2.

What is the difference between VAPP and cVDPV? VAPP is paralysis in a vaccinee or contact caused by the OPV strain itself (very rare); cVDPV arises when the vaccine virus circulates for a long time in a poorly immunised population and regains neurovirulence. Both are reasons for high coverage and adding IPV.

How would you investigate a cluster of abscesses after one outreach session? Stop the session, treat the children, report to the district within 24 hours, keep the vials and syringes, and suspect an immunisation error: contaminated multi-dose vial, reused syringe, wrong technique.

SELF-TEST

Flashcards

28 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Vaccines given at birth in Uganda (current schedule)	**BCG**, **OPV0** and **hepatitis B birth dose** (HepB-BD introduced 2025).
2	Vaccines at 6, 10 and 14 weeks (Uganda)	**OPV**, **DPT-HepB-Hib (pentavalent)**, **PCV** and **rotavirus** at each visit; **IPV** at 6 and 14 weeks (UCG 2023). The current MoH poster shows rotavirus at 6, 10 and 14 weeks.
3	Uganda malaria vaccine: product, schedule, coverage area	**R21/Matrix-M**, 4 doses at **6, 7, 8 and 18 months**, introduced **April 2025** in **107 high-burden districts** (MoH, January 2026).
4	Vaccines at 9 months	**Measles–rubella 1** (0.5 mL SC, left upper arm) + **yellow fever** (0.5 mL, right upper arm).
5	Vaccines at 18 months	**MR2**, **malaria vaccine dose 4** and **vitamin A**.
6	HPV vaccine in Uganda (2025 onward)	Girls aged **10 years** (Primary 4): **single dose**; immunocompromised girls (e.g. HIV) still get **2 doses**.
7	BCG: dose, route, site	**0.05 mL intradermal** (under 12 months; 0.1 mL from 12 months), **right upper arm**.
8	Where are pentavalent and PCV injected in an infant?	Pentavalent: **left** outer thigh. PCV: **right** outer thigh (IPV also right thigh, **2.5 cm** from PCV). All 0.5 mL IM.
9	Storage temperature for vaccines at a health facility	**+2 to +8°C**, with the temperature recorded **twice daily** (including weekends).
10	Vaccines that must never be frozen	**Pentavalent, PCV, IPV, HepB, HPV, Td, rotavirus, malaria vaccine** (and diluents). Only OPV and MR may be stored frozen, at district/national level.
11	How do you read a vaccine vial monitor (VVM)?	Inner square **lighter** than the outer circle = usable (stages 1–2). Square **same as or darker** than circle = **discard** (stages 3–4).
12	The shake test	Tests freeze damage of adsorbed vaccines: shake a suspect vial and a deliberately frozen control; if the suspect sediments **as fast as or faster** than the control, it was frozen: discard.
13	How long can reconstituted BCG, MR and yellow fever be used?	**6 hours** after reconstitution or the **end of the session**, whichever comes first.
14	Multi-dose vial policy for opened OPV, IPV, pentavalent, PCV, Td	May be kept for later sessions up to **4 weeks (28 days)** if not expired, VVM OK, stored at +2 to +8°C, not contaminated and **not opened at outreach** (UCG 2023).
15	True contraindications to vaccination (the short list)	**Anaphylaxis** to a previous dose/component; **live vaccines** (BCG, YF) in symptomatic HIV/severe immunosuppression; **encephalopathy within 7 days** of a pertussis dose; **intussusception** (rotavirus); YF under 6 months.
16	Five false contraindications	Mild illness/low-grade fever, **malnutrition**, **prematurity/LBW**, on antibiotics, breastfeeding, HIV exposure, jaundice, family history of convulsions, previous local reaction.
17	UCG 2023: what to do with a severely ill child who is due vaccines	**Admit and treat**, then **vaccinate at discharge**. Minor illness is not a contraindication.
18	Anaphylaxis after vaccination: first drug and dose	**Adrenaline 1 mg/mL (1:1000) 0.01 mL/kg IM** in the anterolateral thigh (max 0.5 mL), repeat every **5 minutes** as needed.

#	Question	Answer
19	How long should you observe a child after vaccination?	**15–30 minutes**.
20	WHO causality categories for AEFI	**Vaccine product-related**, **vaccine quality defect**, **immunisation error-related**, **immunisation anxiety-related**, **coincidental**.
21	When must a serious AEFI be reported in Uganda, and to whom?	To the **district** (DHO/EPI focal person) **within 24 hours** on the AEFI form; it feeds into UNEPI and the **National Drug Authority** pharmacovigilance system. Keep the vials.
22	Catch-up rules	**Never restart** a series; give all due antigens at first contact; **4 weeks** minimum between doses; live injectables same day or 4 weeks apart.
23	MR in a measles outbreak for a 6-month-old	Give MR from **6 months** ('MR0'); it does **not count**, so repeat at **9 and 18 months**.
24	Td schedule for women of childbearing age	Td1 first contact; Td2 ≥4 weeks later; Td3 6 months later; Td4 1 year later; Td5 1 year later (5 doses = long-term protection).
25	Vitamin A supplementation doses	6–11 months **100,000 IU**; 12–59 months **200,000 IU** every **6 months**.
26	Vaccinating a preterm baby	Use **chronological age** (do not correct). BCG, OPV0 and HepB at birth or discharge.
27	Why is measles vaccine given at 9 months?	**Maternal antibody** neutralises the live vaccine earlier; by 9 months most have lost it while measles risk is high. The 18-month dose catches the ~15% who did not seroconvert.
28	Acute flaccid paralysis surveillance samples	**Two stool samples 24 hours apart within 14 days** of paralysis onset, in any child under 15.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 6-week-old girl (4.2 kg) is brought to a HC III for her first immunisations. She has had a runny nose and a cough for 2 days; temperature 37.8°C, RR 44/min, no chest indrawing, feeding well. Her mother asks whether to come back next week. What is the MOST appropriate action?

A Postpone all vaccines until she has been well for 2 weeks
B Give OPV1, pentavalent 1, PCV1, rotavirus 1 and IPV1 today and advise on fever care
C Give only the oral vaccines today and the injections next week
D Give amoxicillin and vaccinate after the course is complete
E Admit her for observation and vaccinate at discharge

2. A 9-month-old boy (7.0 kg, WHZ −3.1, no oedema) is admitted with severe acute malnutrition and no complications other than poor appetite. His card shows all doses up to 14 weeks. Which plan for his vaccines is MOST appropriate?

A Withhold all vaccines until he reaches −1 Z
B Give MR only after the RUTF course is completed
C Give yellow fever only, because MR is contraindicated in malnutrition
D Give MR1 and yellow fever (and malaria vaccine if in an eligible district) during this admission, before discharge
E Give a double dose of MR because malnourished children respond poorly

3. At the end of a Monday outreach session, the nurse finds that the HC III vaccine fridge thermometer reads −3°C and the pentavalent vials look slightly cloudy. What is the BEST next step before using the pentavalent stock?

A Perform a shake test comparing a suspect vial with a deliberately frozen control vial
B Use the vials because pentavalent is heat-sensitive, not cold-sensitive
C Check the VVM; if it is at stage 1 the vials are safe to use
D Warm the vials to room temperature and use them the same day
E Discard all vaccines in the fridge, including OPV and MR

4. Two minutes after receiving pentavalent, PCV and IPV at 10 weeks, a 5.0 kg baby becomes pale and floppy, with widespread urticaria, stridor and a weak, fast pulse. What is the MOST appropriate first drug?

A Chlorphenamine 0.1 mg/kg IV
B Hydrocortisone 4 mg/kg IV
C Adrenaline 1 mg/mL, 0.5 mL subcutaneously
D Salbutamol nebuliser 2.5 mg
E Adrenaline 1 mg/mL, 0.05 mL (0.05 mg) IM into the anterolateral thigh

5. A 2-year-old girl with symptomatic HIV (oral thrush, WHO stage 3) and no BCG scar is brought for catch-up. Her card shows no BCG and no MR. Which plan is MOST appropriate?

A Give BCG and MR today
B Give BCG only; MR is contraindicated in HIV
C Give MR today; do not give BCG
D Give neither until her CD4 has recovered on ART
E Give BCG at double dose because immunosuppressed children respond poorly

6. A 14-month-old boy from Mukono (9.0 kg) attends clinic. His card shows BCG, OPV0 and the 6-week doses only. His mother is worried that he is 'too old' and 'must start again'. Which plan is MOST appropriate?

A Give OPV2, pentavalent 2, PCV2, IPV, MR1 and yellow fever today, and continue the series at 4-week intervals

B Restart the full infant series with OPV1, pentavalent 1 and PCV1

C Give only MR and yellow fever; the infant vaccines are no longer needed at 14 months

D Give one vaccine per visit to avoid overloading the immune system

E Refer him to the district hospital for a specialist catch-up plan

7. A 4-month-old boy received his 14-week rotavirus vaccine 5 days ago. He now has intermittent screaming episodes with drawing up of the legs, vomiting and a 'red-currant jelly' stool. He is pale between episodes. What is the MOST appropriate next step? Advanced

A Reassure: this is the expected mild gastrointestinal reaction to rotavirus vaccine

B Start oral rehydration salts and zinc for rotavirus gastroenteritis

C Resuscitate, arrange urgent abdominal ultrasound and surgical review for intussusception, and report an AEFI

D Give metronidazole for amoebic dysentery

E Give the next rotavirus dose early to complete the schedule

8. A district EPI officer reports that 4 of 12 infants vaccinated with BCG at one outreach post developed fever and fluctuant abscesses at the injection site over the next week. Several were also given MR from the same session. What is the MOST likely category of these AEFIs? Advanced

A Vaccine product-related reaction

B Coincidental events

C Immunisation anxiety-related reaction

D Vaccine quality defect from the manufacturer

E Immunisation error-related reaction

9. A mother in Lira (a malaria vaccine district) brings her 10-month-old daughter (7.8 kg). The card shows all doses to 14 weeks and MR1 and yellow fever at 9 months, but no malaria vaccine. What is the MOST appropriate advice?

A She is too old for the malaria vaccine; continue nets only

B Give malaria vaccine dose 1 today, then doses 2 and 3 at 4-week intervals and dose 4 about 12 months after dose 3

C Give all three primary malaria doses today in different thighs

D Give malaria dose 1 today and dose 4 at 18 months, skipping doses 2 and 3

E Wait until the next ITN campaign; the vaccine replaces the net

10. A 6-month-old girl in Kasese is a household contact of her 4-year-old brother, who has a confirmed measles rash that appeared yesterday. She has had all vaccines due up to 14 weeks and is well. What is the MOST appropriate action regarding measles vaccination?

A No vaccine: she is too young for MR and is protected by maternal antibody

B Give MR now and count it as MR1; give MR2 at 18 months

C Give vitamin A only

D Give MR now (an extra early dose that does not count), then MR1 at 9 months and MR2 at 18 months

E Give MR at 9 months as usual; exposure does not change the schedule

11. A 10-year-old girl in Primary 4 is being vaccinated against HPV at school in 2026. She is HIV-positive on TLD and suppressed. How many HPV doses does she need according to current Uganda practice?

A One dose, like all other girls
B None, because HPV vaccine is contraindicated in HIV
C Three doses at 0, 1–2 and 6 months
D Two doses, because she is immunocompromised
E One dose now and a booster at 15 years

12. A 3-month-old baby received pentavalent 2 at 10 weeks. Within 24 hours she developed fever of 38.5°C and inconsolable crying for 3 hours, which settled with paracetamol. Her mother asks whether the 14-week dose should be skipped. What is the BEST advice?

A Skip all further pertussis-containing doses
B Give the 14-week dose on schedule; give paracetamol 15 mg/kg if fever recurs
C Give the 14-week dose only after a neurological review
D Replace pentavalent with tetanus toxoid only
E Delay the dose until 6 months, when reactions are fewer

13. At a HC IV, a nurse is preparing measles–rubella vaccine for the afternoon session. Which practice is CORRECT?

A Reconstitute with water for injection if the MR diluent has run out
B Keep a reconstituted MR vial until the next day's session if it stays in the fridge
C Use the same syringe to reconstitute the vial and to vaccinate the first child
D Use diluent at room temperature so that the injection is less painful
E Use the diluent from the same manufacturer, pre-cooled, and discard the vial 6 hours after reconstitution or at the end of the session

14. A baby is born at 34 weeks' gestation weighing 1.9 kg at Mengo. She is now 6 weeks old, weighs 2.6 kg and is well on the KMC unit. Her mother asks whether she should wait until the baby's 'real due date' to start immunisations. What is the BEST advice?

A Give the 6-week vaccines now, by chronological age, at full dose
B Wait until 6 weeks after the expected date of delivery
C Give half doses because the baby is small
D Give oral vaccines only until she weighs 3 kg
E Give the vaccines but repeat them all at 3 months because preterm babies do not respond

15. Which statement about the vaccine vial monitor (VVM) is correct?

A It changes colour when a vaccine has been frozen
B A VVM at stage 3 means the vaccine should be used first
C It shows cumulative heat exposure; use the vial only if the inner square is lighter than the outer circle
D It indicates the expiry date of the vaccine
E It shows whether an opened vial has been contaminated

Answers & explanations

1. B · Give OPV1, pentavalent 1, PCV1, rotavirus 1 and IPV1 today and advise on fever care

B is correct: a mild upper respiratory infection with low-grade fever is a false contraindication; give all due 6-week vaccines today. A is wrong: delaying exposes her to pertussis, pneumococcus and rotavirus and risks dropout. C is wrong: there is no reason to split the vaccines; splitting creates missed opportunities. D is wrong: she has no pneumonia (RR <50, no indrawing), and antibiotics are not a contraindication anyway. E is wrong: admission and vaccination at discharge are for severely ill children, not a well-feeding baby with a cold.

2. D · Give MR1 and yellow fever (and malaria vaccine if in an eligible district) during this admission, before discharge

D is correct: malnutrition is a false contraindication and malnourished children are at highest risk of dying from measles; give the due 9-month vaccines (and malaria doses if eligible) during the admission. A is wrong: waiting for nutritional recovery leaves him unprotected for weeks to months. B is wrong: it delays MR without reason. C is wrong: MR is not contraindicated in malnutrition. E is wrong: there is no double dose; the standard 0.5 mL dose is used.

3. A · Perform a shake test comparing a suspect vial with a deliberately frozen control vial

A is correct: pentavalent is an adsorbed, freeze-sensitive vaccine; the shake test detects freeze damage (the suspect vial sediments as fast as the frozen control). B is wrong: freezing destroys adsorbed vaccines (pentavalent, PCV, HepB, Td, IPV, HPV). C is wrong: the VVM shows heat exposure only; it says nothing about freezing. D is wrong: thawing does not restore a freeze-damaged vaccine. E is wrong: OPV and MR are not damaged by freezing; discard only what fails the shake test (and report the cold-chain break).

4. E · Adrenaline 1 mg/mL, 0.05 mL (0.05 mg) IM into the anterolateral thigh

E is correct: this is anaphylaxis; the first-line drug is IM adrenaline 0.01 mg/kg (0.01 mL/kg of 1 mg/mL), here 0.05 mL, into the thigh, repeated every 5 minutes if needed. A is wrong: antihistamines are adjuncts and do not treat airway obstruction or shock. B is wrong: steroids act too slowly and are adjuncts only. C is wrong: 0.5 mL is the adult maximum (ten times this infant's dose), and the subcutaneous route is absorbed less reliably than IM. D is wrong: salbutamol does not treat laryngeal oedema or shock.

5. C · Give MR today; do not give BCG

C is correct: UCG 2023 says do not give a missed BCG to a child with symptomatic HIV (risk of disseminated BCG), but give measles–rubella even in symptomatic HIV because measles is so dangerous. A is wrong: BCG, a live vaccine, is contraindicated in symptomatic HIV. B is wrong: it reverses the rule; MR is given. D is wrong: delaying MR leaves her exposed to measles, which kills HIV-positive children. E is wrong: increasing a live-vaccine dose in immunosuppression is dangerous.

6. A · Give OPV2, pentavalent 2, PCV2, IPV, MR1 and yellow fever today, and continue the series at 4-week intervals

A is correct: never restart a series; count documented doses, give all due antigens at the first contact (different sites), and continue at minimum 4-week intervals. B is wrong: restarting wastes doses and delays protection; doses already given count. C is wrong: he still needs pentavalent (diphtheria, pertussis, tetanus, HepB, Hib), PCV and polio doses at 14 months. D is wrong: multiple vaccines at one visit are safe and reduce missed opportunities. E is wrong: catch-up is routine work at any immunising facility.

7. C · Resuscitate, arrange urgent abdominal ultrasound and surgical review for intussusception, and report an AEFI

C is correct: colicky pain, vomiting and red-currant stool within 7 days of rotavirus vaccine suggest intussusception, a rare serious AEFI; resuscitate, confirm by ultrasound, reduce (air/hydrostatic) or operate, report, and give no further rotavirus doses. A is wrong: the expected reaction is mild, short-lived GI upset, not bloody stools and colic. B is wrong: treating as gastroenteritis delays a surgical emergency. D is wrong: amoebic dysentery is rare in young infants and this picture is classic intussusception. E is wrong: a history of intussusception is a contraindication to further rotavirus doses.

8. E · Immunisation error-related reaction

E is correct: a cluster of hot, fluctuant abscesses after one session suggests non-sterile technique or a contaminated reconstituted vial (immunisation error): for example, keeping reconstituted BCG beyond 6 hours or reusing syringes. A is wrong: the expected BCG reaction is a small papule that ulcerates over weeks, not a cluster of febrile abscesses within a week. B is wrong: a cluster linked to one session is unlikely to be chance. C is wrong: anxiety reactions are fainting or hyperventilation in older children, not abscesses. D is wrong: a manufacturing defect would affect many sites using the same lot, not one outreach post.

9. B · Give malaria vaccine dose 1 today, then doses 2 and 3 at 4-week intervals and dose 4 about 12 months after dose 3

B is correct: the malaria vaccine is for young children in eligible districts; a late starter begins now and completes the 3 primary doses at minimum 4-week intervals, with the booster about a year after dose 3. A is wrong: 10 months is within the target age group (under 2). C is wrong: doses of the same vaccine need 4-week intervals to build immunity. D is wrong: skipping primary doses leaves her unprotected. E is wrong: the vaccine adds to nets; it never replaces them.

10. D · Give MR now (an extra early dose that does not count), then MR1 at 9 months and MR2 at 18 months

D is correct: UCG 2023 allows MR from 6 months for an exposed child; given early (within 72 hours of exposure it may prevent disease), it does not count, so MR is repeated at 9 and 18 months. A is wrong: maternal antibody is waning by 6 months and exposure is intense in a household. B is wrong: a dose before 9 months gives a weaker response and must not count. C is wrong: vitamin A is for children who have measles (treatment), not a substitute for vaccination of contacts. E is wrong: waiting ignores the outbreak risk.

11. D · Two doses, because she is immunocompromised

D is correct: since 2025 Uganda gives a single dose to 10-year-old girls, except immunocompromised girls (such as those with HIV), who continue with two doses (MoH January 2026). A is wrong: the single-dose schedule excludes immunocompromised girls. B is wrong: HPV vaccine is not live and is safe in HIV; girls with HIV are at high risk of cervical cancer. C is wrong: three doses are used in some countries for immunocompromised people, but Uganda's current guidance is two. E is wrong: no booster at 15 years is in the Ugandan schedule.

12. B · Give the 14-week dose on schedule; give paracetamol 15 mg/kg if fever recurs

B is correct: fever and a few hours of crying after a whole-cell pertussis vaccine are expected minor reactions, not contraindications; continue on schedule with paracetamol advice. A is wrong: only anaphylaxis or encephalopathy within 7 days stops further pertussis doses. C is wrong: no neurological review is needed for an expected febrile reaction. D is wrong: removing the pertussis, Hib and HepB components would leave her unprotected. E is wrong: delay prolongs the period of pertussis risk, which is greatest in young infants.

13. E · Use the diluent from the same manufacturer, pre-cooled, and discard the vial 6 hours after reconstitution or at the end of the session

E is correct: UCG 2023 requires each vaccine's own pre-cooled diluent, and reconstituted MR (like BCG and YF) is discarded after 6 hours or at the session's end, whichever comes first. A is wrong: UCG says never use water for injection as a diluent; wrong diluents have caused deaths. B is wrong: reconstituted MR has no preservative and must not be kept overnight (risk of contamination and toxic shock). C is wrong: a separate mixing syringe is used and discarded; every child gets a new AD syringe. D is wrong: warm diluent added to the cold vaccine heat-shocks it; diluent must be cooled.

14. A · Give the 6-week vaccines now, by chronological age, at full dose

A is correct: preterm and LBW babies are vaccinated according to chronological age with full doses; prematurity is a false contraindication. B is wrong: correcting for prematurity delays protection when infections are most dangerous. C is wrong: doses are never reduced for size. D is wrong: there is no weight threshold for injectable vaccines. E is wrong: preterm infants respond adequately; repeating doses is unnecessary (the exception is a HepB birth dose given under 2 kg, which does not count towards the series).

15. C · It shows cumulative heat exposure; use the vial only if the inner square is lighter than the outer circle

C is correct: the VVM records cumulative heat exposure; usable while the inner square is lighter than the circle (stages 1–2), discard at stages 3–4. A is wrong: the VVM does not detect freezing (use the shake test or freeze indicators). B is wrong: stage 3 is the discard point; 'use first' applies to stage 2. D is wrong: expiry is printed separately and must also be checked. E is wrong: contamination cannot be seen on the VVM; MDVP rules govern opened vials.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The child with no card and a measles outbreak

During a measles outbreak in Kampala, a 13-month-old boy is brought to Mengo's outpatient clinic with a 3-day cough and runny nose; temperature 37.9°C, RR 38/min, no rash, alert and feeding. Weight 8.3 kg (WAZ −1.8), MUAC 13.0 cm. His mother lost his card; she remembers 'injections when he was small' but no visits after 3 months. His 5-year-old sister also has no card.

1. Can he be vaccinated today? Justify.

> **Yes.** Mild URTI and low-grade fever are false contraindications; there are no danger signs and he is feeding. In an outbreak, delaying MR is dangerous.

2. How will you establish what he has had?

> Ask where he was vaccinated and check that facility's **child register**; look for a **BCG scar** (right arm). If no record is found, **assume he is unvaccinated** for doses that cannot be confirmed: extra doses are safe.

3. He has a BCG scar; the register confirms BCG, OPV0 and 6- and 10-week doses. What do you give today?

> **OPV3, pentavalent 3, PCV3, IPV (2nd dose if only one given), MR1, yellow fever** (live vaccines same day), plus **malaria vaccine** if the district is eligible and **vitamin A 200,000 IU**. Use separate sites; record on a **new card**.

4. What is the plan after today?

> **MR2** at 18 months (at least 4 weeks after MR1), remaining malaria doses at 4-week intervals then a booster, vitamin A every 6 months, growth monitoring. Write the next date on the card and link to a VHT for follow-up.

5. What about his 5-year-old sister?

> Check her history the same way; give **MR** (any unvaccinated child), catch-up **Td/pentavalent-type** doses per national guidance (Hib not usually needed after 5 years; PCV catch-up generally not after 5), and OPV/IPV. Every contact is an opportunity, including siblings.

6. Two days later the boy develops a maculopapular rash, conjunctivitis and fever 39°C. What do you do?

> This is likely **measles infection** (incubated before vaccination; the vaccine rash comes at 7–12 days and is mild). Isolate, notify the district (**measles IgM** sample), give **vitamin A** (200,000 IU days 1 and 2, repeat at 2–4 weeks if eye signs), screen for pneumonia, diarrhoea and malnutrition, and vaccinate contacts. Report as an AEFI only if the investigation suggests a vaccine link.

Take-away: Minor illness and a lost card are not reasons to send a child away: confirm what you can, assume unvaccinated where you cannot, give everything due at once, never restart, and immunise the siblings too.

Case 2: Anaphylaxis at an outreach session

You are a clerk helping at an immunisation outreach under a tree in Wakiso. A 10-year-old girl (28 kg) receives HPV vaccine. Five minutes later she becomes anxious, says her throat feels tight, and develops generalised urticaria and wheeze; pulse 140/min, weak.

1. How do you distinguish anaphylaxis from an immunisation anxiety reaction (fainting)?

> **Fainting**: pallor, sweating, slow pulse, brief loss of consciousness, quick recovery lying flat, **no** urticaria, stridor or wheeze. **Anaphylaxis**: skin signs (urticaria, angio-oedema) plus **airway, breathing or circulation** involvement (stridor, wheeze, tachycardia, hypotension). This girl has anaphylaxis.

2. What is your immediate management, with doses?

> Call for help, lie her down (or sit up if breathing is easier; legs raised if hypotensive). **Adrenaline 1 mg/mL 0.01 mL/kg = 0.28 mL** (round to 0.3 mL) **IM anterolateral thigh**; repeat every **5 minutes** if no improvement. Oxygen if available, IV access and **normal saline 20 mL/kg (about 560 mL)** if shocked.

3. She improves after one dose. What next?

> Adjuncts: **chlorphenamine 0.1 mg/kg** (2.5–4 mg) for itch; consider **hydrocortisone 4 mg/kg** (about 110 mg). **Transfer** to the nearest hospital for observation for at least **6–12 hours** (biphasic reactions), with a health worker and a second adrenaline dose drawn up.

4. What must happen administratively?

> Fill in the **AEFI reporting form** and notify the **district within 24 hours** (serious AEFI); keep the **vial and lot number**; record on her card that she must **not receive this vaccine again**; inform her parents honestly.

5. Other girls in the queue start to feel faint and dizzy. What is happening and what do you do?

> Likely **immunisation anxiety-related reactions** (mass psychogenic response), common in adolescent school campaigns. Move girls away from the scene, vaccinate seated, lie down anyone who feels faint, reassure calmly, and examine each for objective signs of anaphylaxis before attributing symptoms to anxiety.

6. What should have been in place before the session?

> **Emergency tray** (adrenaline 1 mg/mL ×3, 1 mL syringes, needles, chart of doses), trained staff, a **15–30-minute observation area**, a referral plan and transport, a phone with the district contact.

> **Take-away:** Anaphylaxis is rare but must be expected at every session: recognise airway, breathing or circulation involvement, give IM adrenaline 0.01 mL/kg of 1 mg/mL into the thigh at once, observe, refer and report within 24 hours.

Case 3: Is it the vaccine? Fever and convulsion after 9-month vaccines

A 9-month-old girl from Mityana (7.9 kg) received MR1, yellow fever and malaria dose 3 eight days ago. She is brought to the HC IV with fever 39.6°C for one day and a generalised convulsion lasting 3 minutes, which stopped on its own. She is now drowsy but rousable, RR 40/min, no neck stiffness, fontanelle flat.

1. What are your immediate actions?

> ABC; recovery position; **RBS**; temperature control (paracetamol **15 mg/kg ≈ 120 mg**); if she convulses again for >5 minutes, **diazepam 0.5 mg/kg rectally (4 mg)**. Assess for danger signs and shock.

2. List your differential diagnosis.

> (1) **Febrile seizure** due to post-MR fever (days 5–12) or due to another infection; (2) **severe malaria/cerebral malaria**; (3) **bacterial meningitis**; (4) other infections (pneumonia, UTI, otitis); (5) hypoglycaemia; (6) very rarely, YF vaccine-associated neurological disease.

3. Which investigations will you do, and why?

> **Malaria RDT/blood smear** (coincidental malaria is common, and the malaria vaccine does not give full protection); **RBS**; **FBC**; **urinalysis**; **lumbar puncture** if she does not return to full consciousness within an hour, has meningeal signs, or is under 12 months with a first seizure and an unclear source (many clinicians do an LP in this age group).

4. RDT is positive and she recovers fully within 45 minutes. How do you manage and classify?

> Treat **malaria** (uncomplicated: artemether–lumefantrine by weight; if any severe feature, IV artesunate 3 mg/kg for <20 kg). The seizure is most likely a **febrile seizure due to coincidental malaria**. If an AEFI report is made (seizure within the post-MR window), the investigation would likely classify it as **coincidental**.

5. Her mother asks whether her daughter should receive MR2 and malaria dose 4. What do you advise?

> **Yes.** A febrile seizure is not a contraindication to further MR or malaria doses. Explain fever care after vaccination and that measles itself causes far more seizures, encephalitis and deaths than the vaccine.

6. How does this case affect public trust, and what is your role?

> Rumours that 'the new malaria vaccine causes fits' can reduce uptake (already a problem with completion). Communicate clearly, document the cause, report through the AEFI system, and reassure the community via VHTs.

> **Take-away:** Most events after vaccination are coincidental: look for malaria, meningitis and hypoglycaemia first, treat the cause, report and classify honestly, and keep the child on schedule.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.